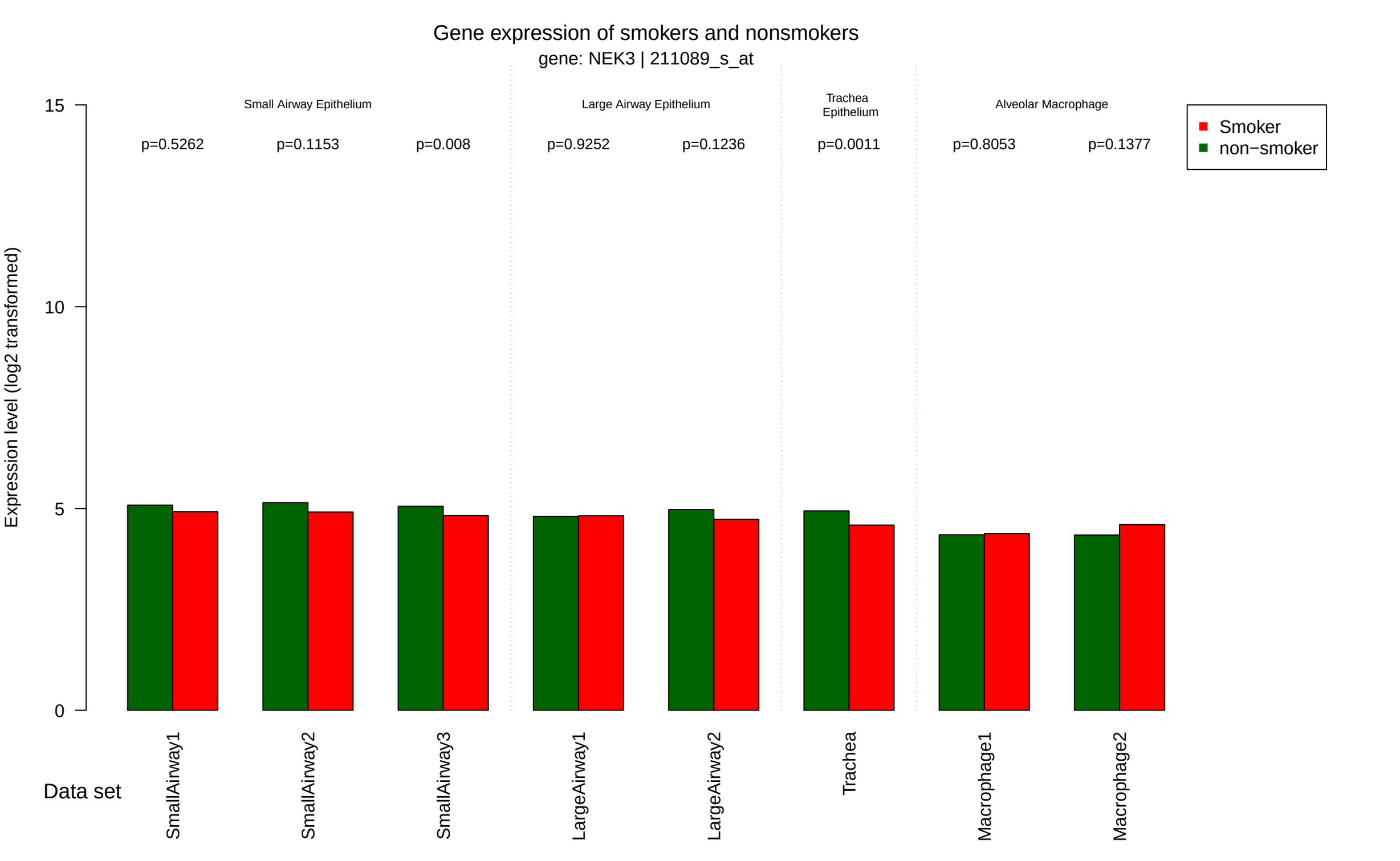
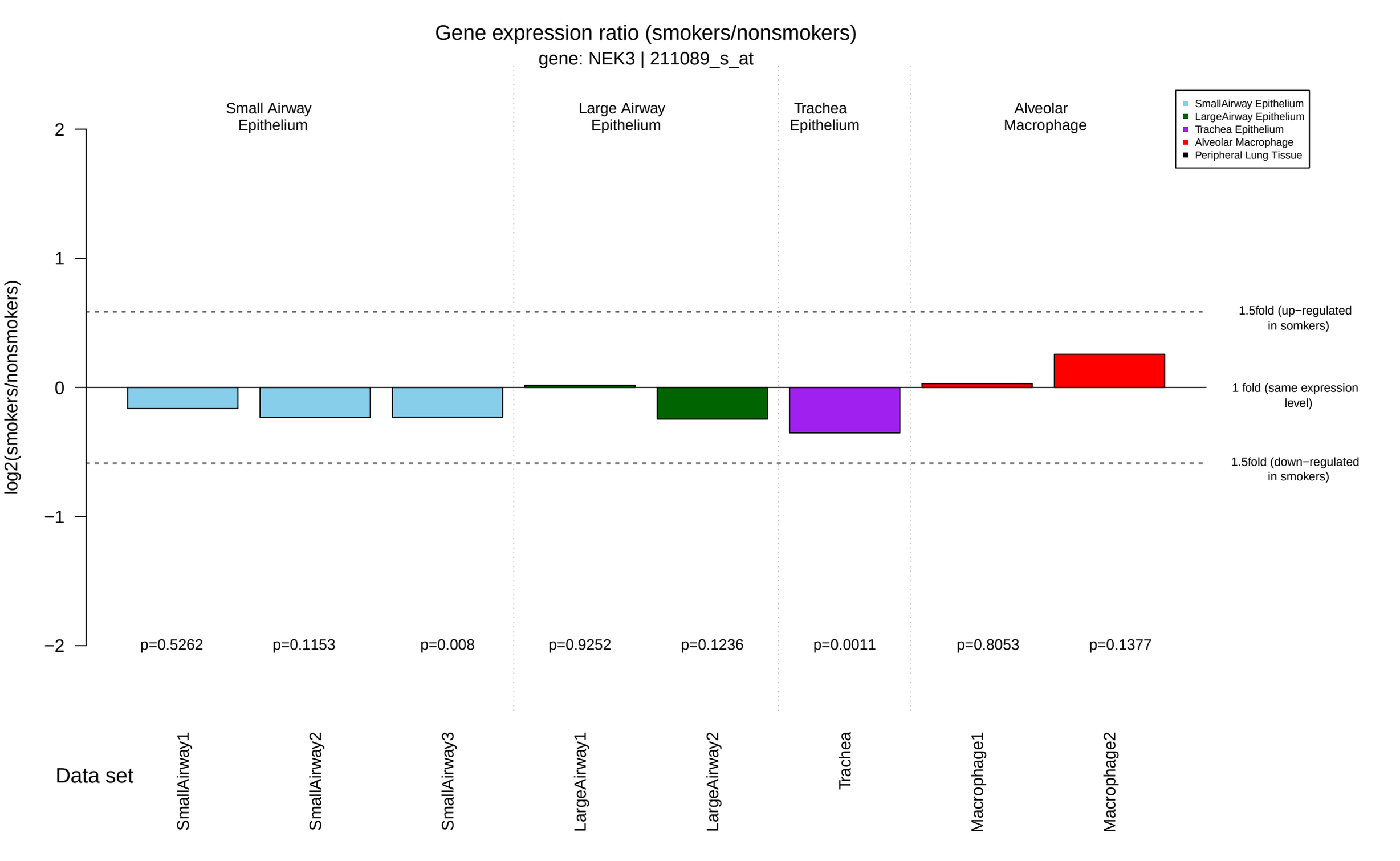
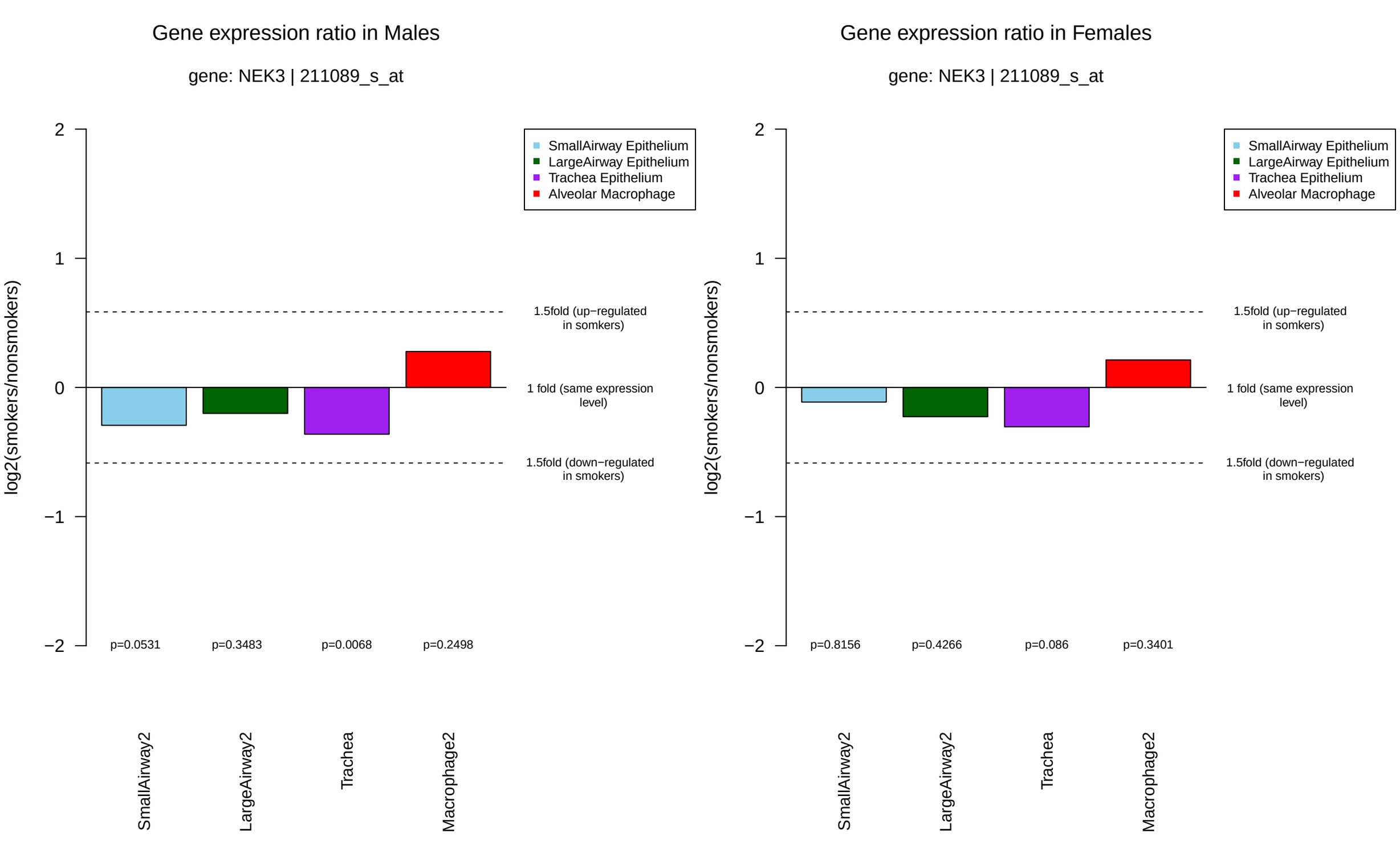
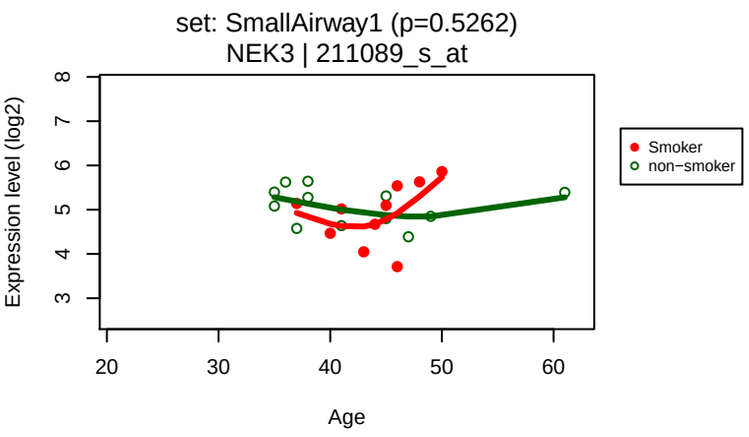
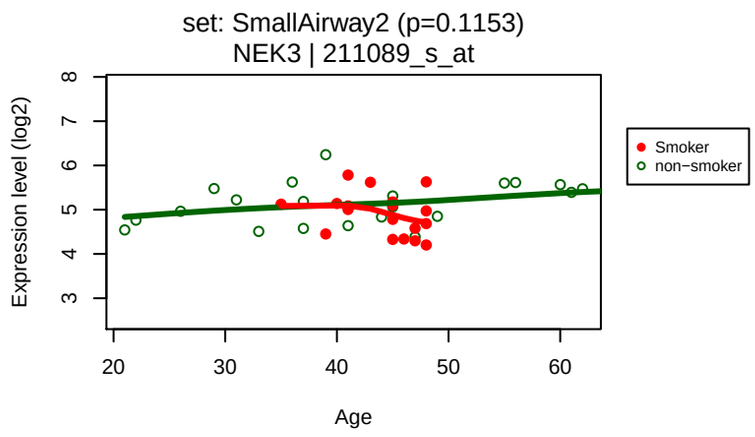
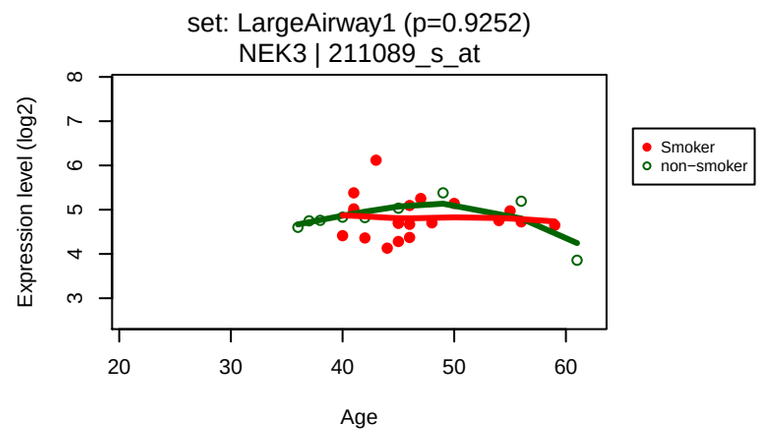

Gene expression of smokers and nonsmokers
gene: NEK3 | 211089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5262
p=0.1153
p=0.008
p=0.9252
p=0.1236
p=0.0011
p=0.8053
p=0.1377
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

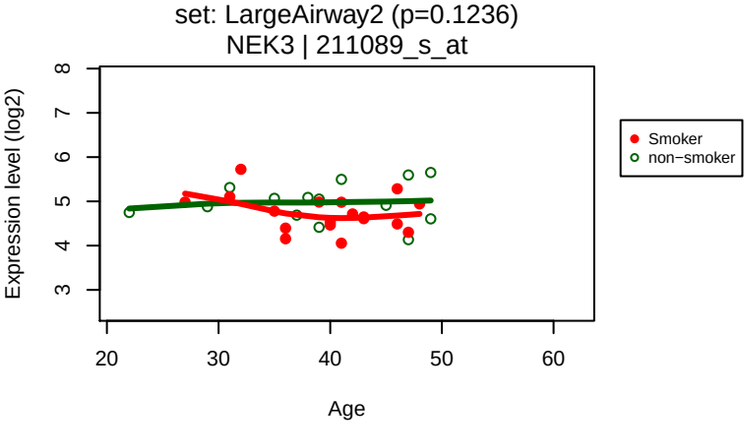

Gene expression ratio (smokers/nonsmokers)
gene: NEK3 | 211089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5262
p=0.1153
p=0.008
p=0.9252
p=0.1236
p=0.0011
p=0.8053
p=0.1377
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

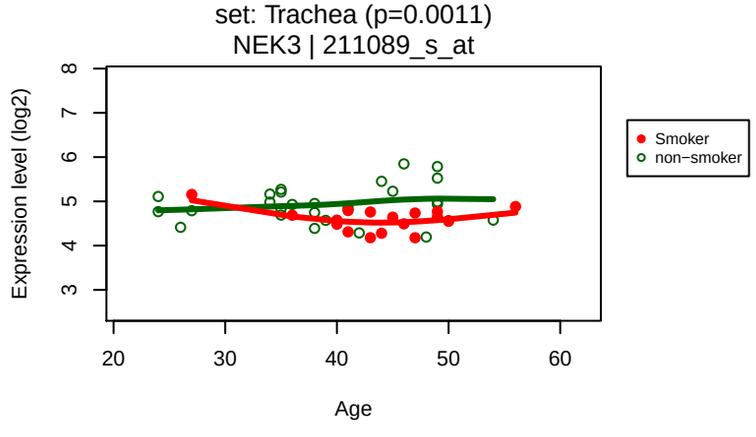

Gene expression ratio in Males
gene: NEK3 | 211089_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0531
p=0.3483
p=0.0068
p=0.2498
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK3 | 211089_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8156
p=0.4266
p=0.086
p=0.3401
SmallAirway2
LargeAirway2
Trachea
Macrophage2

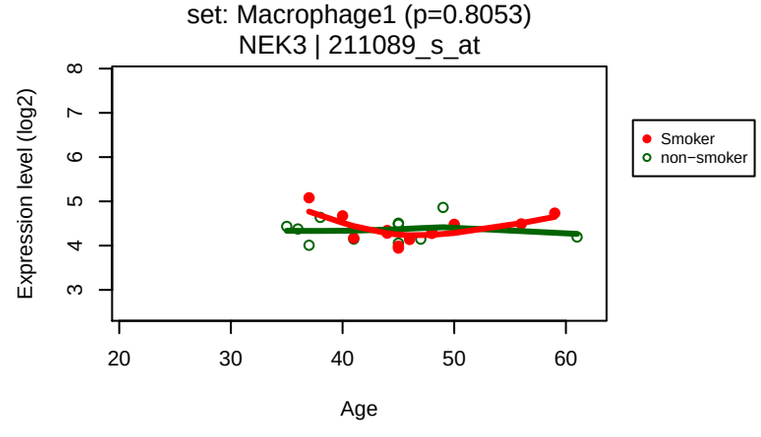

set: SmallAirway1 (p=0.5262)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker

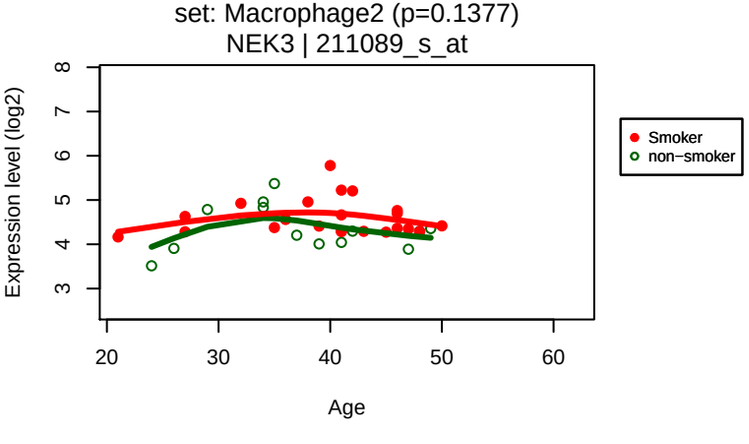

set: SmallAirway2 (p=0.1153)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker

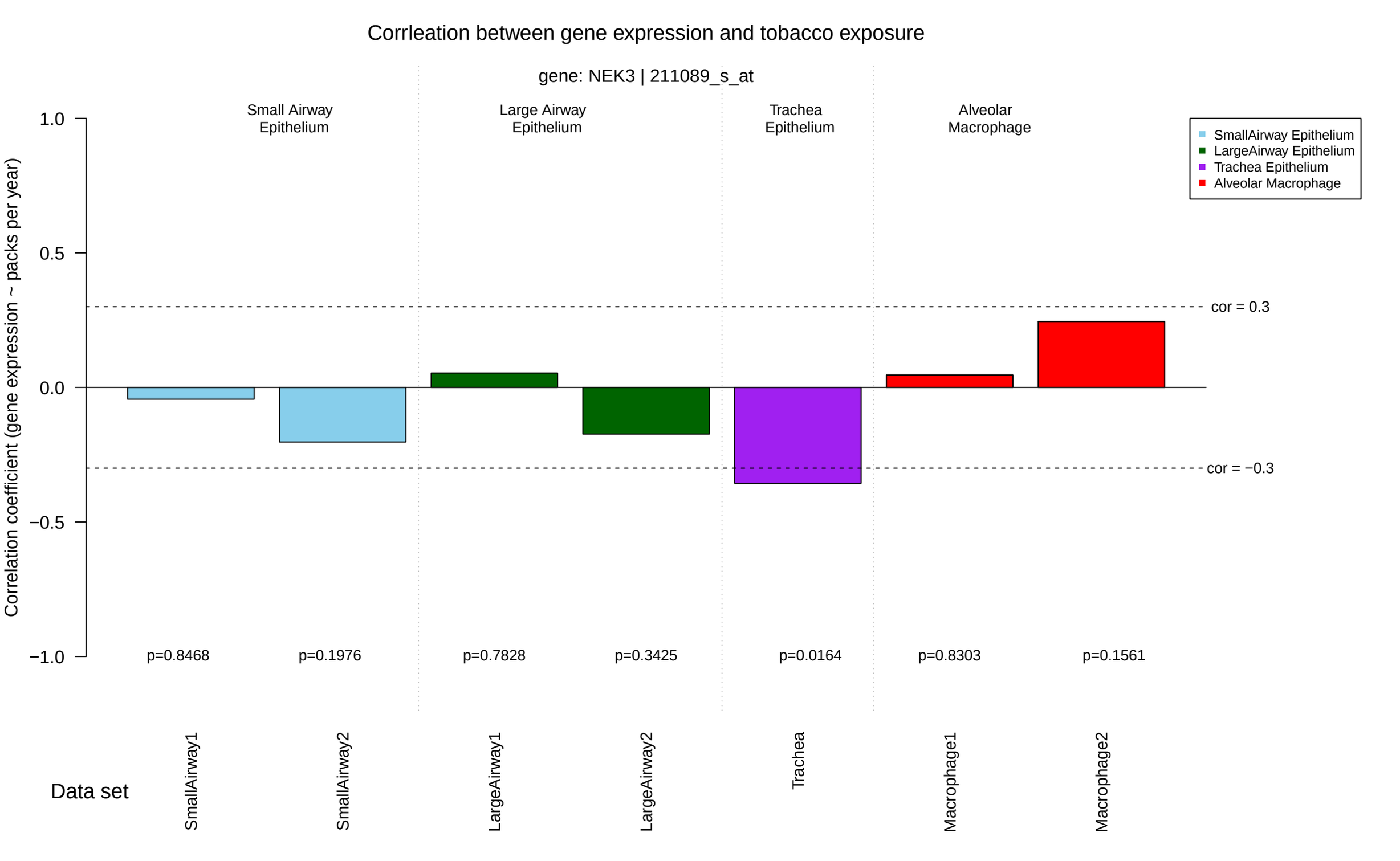

set: LargeAirway1 (p=0.9252)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1236)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8053)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1377)
NEK3 | 211089_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK3 | 211089_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8468
p=0.1976
p=0.7828
p=0.3425
p=0.0164
p=0.8303
p=0.1561
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2